

Chapter 4 – Activity Based Costing Part 1

Traditional approach to allocating Manufacturing overhead

Manufacturing overhead applied to production using a rational approach, systematic manner, based on an allocation base.

Broad averaging of costs

Advantage – Simple, easy method

Disadvantage – Does not accurately reflect the actual amount of overheads consumed by the product.

Increasing accuracy and complexity

Simplest method has only one cost pool and one allocation base plantwide and the same OH rate is used for all products.

Accuracy may be improved by increasing number of cost pools with different allocation bases

Accuracy may also be improved by computing allocation rates separately for each division (instead of plantwide)

Increasing complexity

Example

Consider a plant with two departments – Dept A and Dept B.

Plant-wide OH rate will allocate OH using a single rate (based on a single allocation base) regardless of whether the job is in department A or B.

Accuracy may be improved by using different rates with different cost pools and allocation bases for each department

For instance, plant-wide rate may use direct labor hours (or machine hours, but not both)

Increasing accuracy – Dept A's appropriate base is direct labor hours, while Dept B's appropriate base is machine hours

Example

	Dept A	Dept B	Plantwide
Overhead	\$300,000	\$450,000	\$750,000
Direct labor hours (DLH)	8,000	7,000	15,000
Machine hours (MH)	750	1,200	1950
DLH rate	\$37.50	n/a	\$50.00
MH rate	n/a	\$375	\$384.62

Example

Computing cost of Job 457 which incurs 1000 DLH in Dept A and 1000 DLH in Dept B; and 50 MH in Dept A and 75 MH in Dept B

Using Plantwide DLH rate, the amount of OH allocated to the job is $2000 \times \$50 = \$100,000$

Using Plantwide MH rate, the amount of OH allocated to the job is $125 \times 384.62 = \$48,077.50$

Using departmental rates (DLH for Dept A and MH for department B, the amount of OH allocated is $(1000 \times \$37.50) + (75 \times \$375) = \$65,625$

Example

Different methods give different costs for the product ranging from approx. \$48,000 to \$100,000.

Plantwide OH rates are based on some type of averaging.

The department based costing is likely to be more accurate because each department uses the “best” allocation base for that cost pool.

Using this method of costing is likely to result in “undercosting” or “overcosting”

Undercosting – actual costs are higher than allocated costs

Overcosting – actual costs are lower than allocated costs

Under and over costing

UNDERCOSTING occurs when the cost measurement system reports a cost for a product that is below the cost of the resources the product consumes

OVERCOSTING occurs when the cost measurement system reports a cost for a product that is above the cost of the resources the product consumes.

Using a traditional uniform allocation rate – Cross subsidization

	Drinks	Dinner	Total
Adam	\$25	\$25	\$50
Danielle	\$ 5	\$15	\$20
Tyler	\$30	\$30	\$60
Ryan	\$30	\$40	\$70

\$200

That's \$50 each before
tip, \$60 each should
cover the tip.

Issues with under and over costing

Since total costs are constant, if one product is undercosted, at least one other product is overcosted

This is called cross-subsidization

Undercosting a product makes it seem more profitable than it actually is

Overcosting a product makes it seem less profitable than it actually is

Incorrect decisions

Incorrect costing leads to incorrect decisions

If price is based on cost, the pricing is likely to be incorrect.

Likely to focus on the incorrect product in marketing

Likely to focus on incorrect product in cost cutting efforts

Potential Solution

Is it possible to identify at a more detailed level as to what product consumes what resources?

What activities are demanded by each product?

What resources (costs) are generated by each activity?

Create cost pools for each activity

Identify the allocation base (amount of activity)

Allocate costs based to each product based on the amount of each activity demanded by each product

Limitations of a simple single cost driver overhead allocation system

